Unit: 2
System of Forces

FORCE

It is a physical quantity that changes or tries to change the state of rest or uniform motion of a body. Unit of force is Newton N (*i.e.*, kg m/s^2).

Force can also be defined as an external agency which produces or tends to produce, destroy or tends to destroy the motion. Force is a vector quantity defined completely by its:

- Magnitude
- Direction
- Point of application
- Line of action.

A force or system of forces when acting on a body may:

- Change its state of rest or motion
- Accelerate or retard its motion
- Change its shape or size
- Turn or rotate it, and
- Keep it in equilibrium

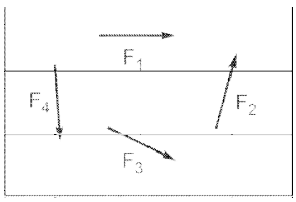
SYSTEM OF FORCES

When a body is acted upon by two or more number of forces simultaneously, it will form a system of forces. Considering the plane in which forces are applied and depending upon the position of line of action, system of forces may be categorized as:

1. Coplanar forces
2. Non-coplanar forces
3. Collinear forces
4. Non-collinear forces
5. Concurrent forces
6. Non-concurrent forces
7. Parallel forces.

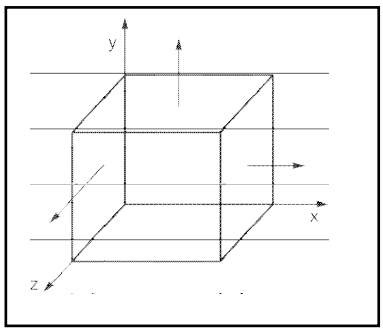
Coplanar Forces

When all the forces acting on a body lie in a same plane, it is termed as coplanar force system. It is a two dimensional force system



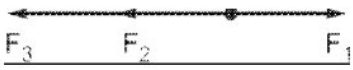
Non-coplanar Forces

When all the forces acting on a particular body cannot pass through a single plane; it is called non-coplanar forces. It is a three dimensional force system.



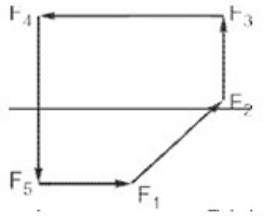
Collinear Forces

In this type of force system the line of action of all forces lie along the same straight line.

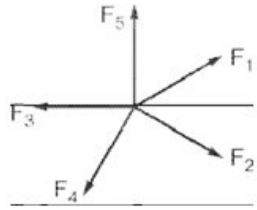


Non-collinear Forces

If line of action of all the forces is not passing through a single straight line, it is called non-collinear force system.



Non-collinear forces.



Concurrent forces.

Concurrent Forces

If lines of action of all the forces are passing through a single common point, it is called as concurrent force system. Point of intersection of all the forces is termed as point of concurrency.

Non-concurrent Forces

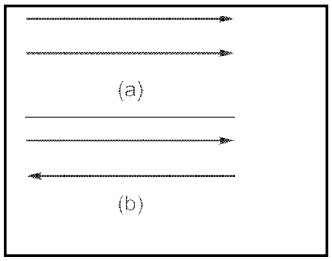
Forces are not passing through a common point in non-concurrent force system.

Parallel Force System

If lines of action of all the forces are parallel to each other, it is called parallel force system. Depending on the direction of forces it may be

(i) **Like parallel forces:** Forces are parallel and in same direction.

(ii) **Unlike parallel forces:** Forces are parallel but in opposite direction.



EQUILIBRIUM, RESULTANT AND EQUILIBRANT

When two or more than two forces acting on a body in such a way that body remains in a state of rest or of uniform motion (no acceleration or retardation), then the system of forces is said to be in equilibrium. Conditions of equilibrium for different system of forces are listed below:

(*a*) Coplanar forces system:

— Concurrent force: $\Sigma F_x = 0$, $\Sigma F_y = 0$

— Non concurrent forces: $\Sigma F_x = 0$, $\Sigma F_y = 0$, $\Sigma M = 0$

(*b*) Non coplanar force system:

— Concurrent forces: $\Sigma F_x = 0$, $\Sigma F_y = 0$, $\Sigma F_z = 0$

— Non concurrent forces: $\Sigma F_x = 0$, $\Sigma F_y = 0$, $\Sigma F_z = 0$, $\Sigma M = 0$

When a body is acted upon by a system of forces, then vectorial sum of all the forces is known as resultant. Hence resultant refers to a single force which produces the same effect as is done by the combined effect of several forces.

A number of forces may act on a body in such a manner, that the body is not in equilibrium. A single force which brings the body in equilibrium is known as equilibrant. Equilibrant is equal and opposite to the resultant of several forces acting on the body.

Parallelogram Law of Forces

This law is used to find out the resultant of two forces acting at a point of a rigid body. It states that "If two forces acting simultaneously on a rigid body, can be represented in magnitude and direction by two adjacent sides of a parallelogram then their resultant may be represented in magnitude and direction by the diagonal of the parallelogram, passing through their point of intersection".

Let P and R are the two force acting at a point A. Parallelogram $ABCD$, as shown in Fig. 1.8, is completed using dotted lines. Drop a perpendicular from C to meet the extended line AB at point E.

Considering ΔACE, we can write

$$AC^2 = CE^2 + AE^2$$

$$AC^2 = CE^2 + (AB + BE)^2$$

$$R^2 = Q^2 \sin^2\theta + (P + Q\cos\theta)^2$$

$$= Q^2 \sin^2\theta + P^2 + Q^2\cos^2\theta + 2PQ\cos\theta$$

$$= P^2 + Q^2(\sin^2\theta + \cos^2\theta) + 2PQ\cos\theta$$

$$= P^2 + Q^2 + 2PQ\cos\theta.$$

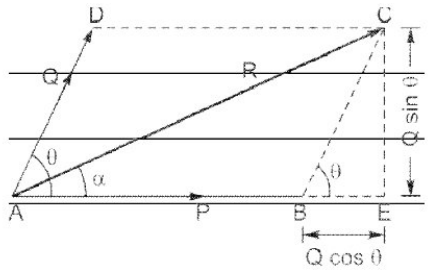


Fig. 1.8

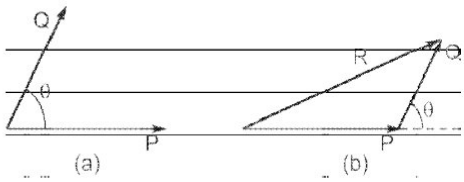
or $$R = \sqrt{P^2 + Q^2 + 2PQ\cos\theta}$$

Angle of resultant R from force P is

$$\alpha = \tan^{-1}\frac{Q\sin\theta}{P + Q\cos\theta}$$

Triangular Law of Forces

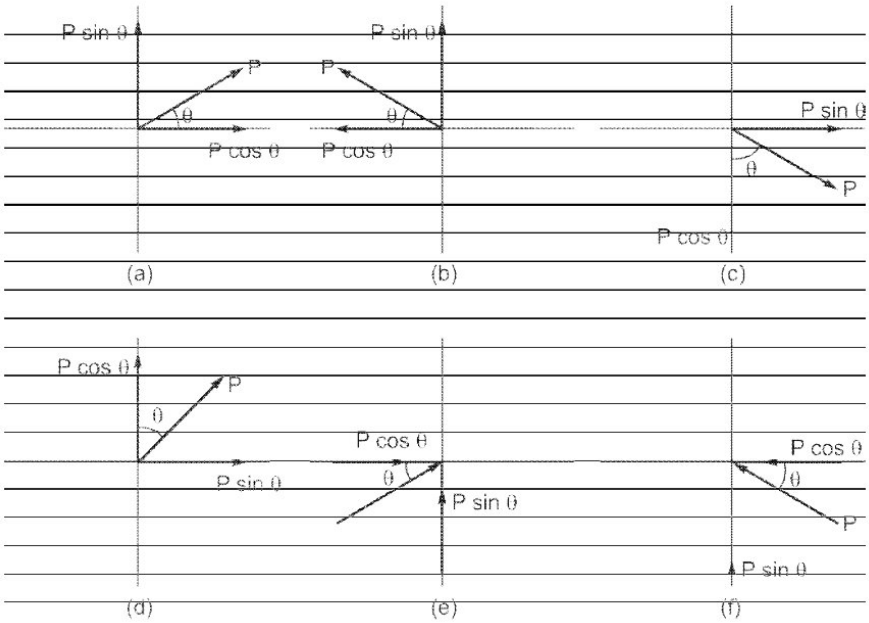
It states that "If two forces acting simultaneously on a body are represented by the sides of a triangle taken in order, then their resultant is represented by the closing side of the triangle taken in opposite order."

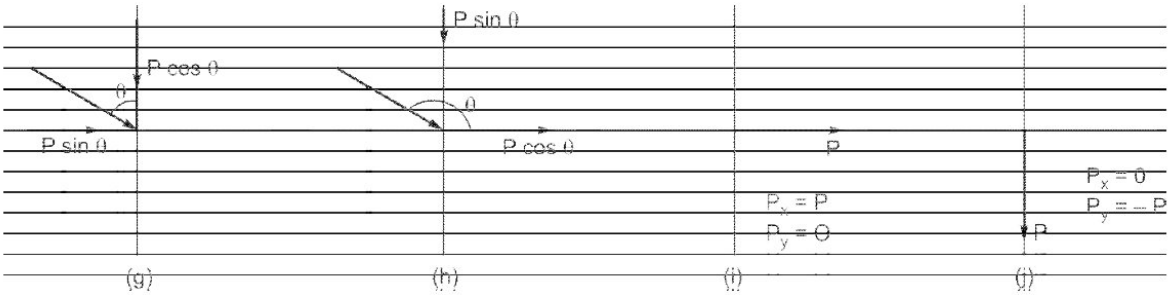


COMPOSITION AND RESOLUTION OF FORCES

Composition or compounding is the procedure to find out single resultant force of a force system, while resolution is the procedure of splitting up a single force into number of components without changing the effect of the same.

In orthogonal resolution a single force is splitted into two mutually perpendicular direction. Some examples are given below:





METHODS OF COMPOSITION

Following steps are involved for finding the resultant for coplanar concurrent force system:

- Resolve all the forces in two mutually perpendicular directions (*i.e.* along horizontal and vertical direction).
- Take upward and rightward forces as positive and downward and leftward forces as negative.
- Find out alzebric sum of all the vertical $[\Sigma F_y]$ and horizontal $[\Sigma F_x]$ components.
- Calculate the magnitude of the resultant force as, $R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2}$
- Its direction can be calculated by

$$\alpha = \tan^{-1}\left[\frac{\Sigma F_y}{\Sigma F_x}\right]$$

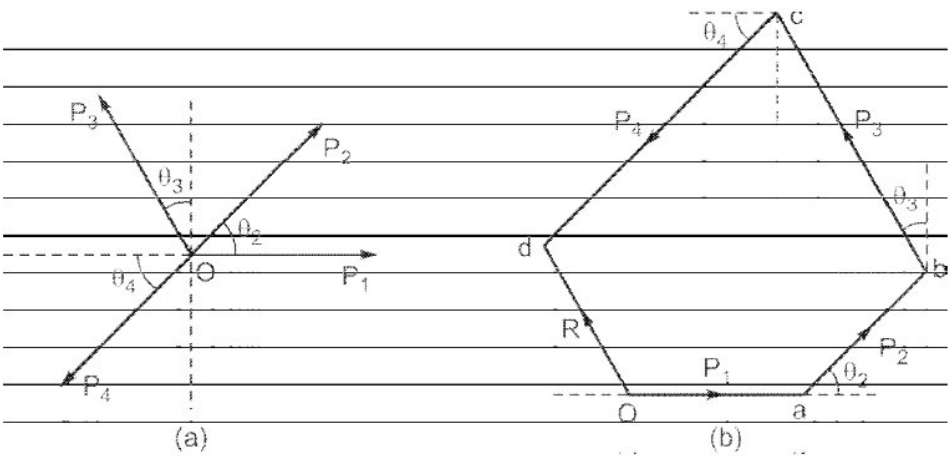
α is the angle of resultant from x-axis.

In case of non-concurrent forces magnitude and direction of force is not sufficient to define resultant. In this case it is required to find out the point of application of resultant force.

Law of Polygon

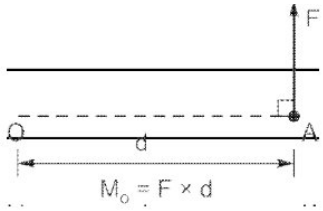
Law of polygon states that "If a number of concurrent forces acting simultaneously on a body are represented in magnitudes and direction by the sides of a polygon, taken in order, then the resultant is represented in magnitude and direction by the closing side of the polygon, taken in opposite order."

Consider forces P_1, P_2, P_3 and P_4 acting on a body at point O as shown in Fig. Draw a line oa to represent P_1, line ab to represent P_2, line bc to represent P_3 and line cd to represent P_4. The polygon is computed by drawing the closing line od. This closing line od represents the resultant of the given system in magnitude, line of action and direction.

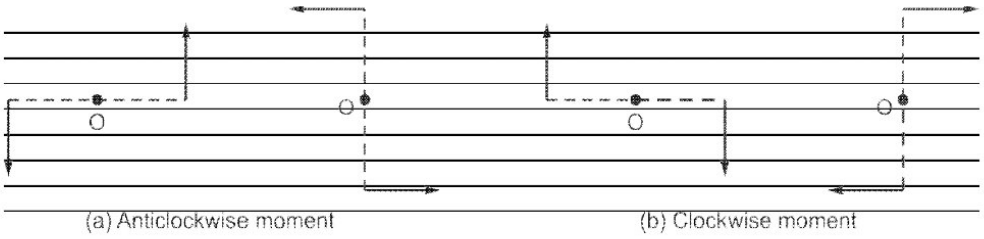


MOMENT OF A FORCE

It is a turning effect produced by a force on the body on which it acts. Moment of a force is equal to the product of the force and the perpendicular distance of the point, from the line of action of force, about which the moment is required.



Moments are considered as positive or negative depending upon its tendency to rotate the body in the clockwise (assuming (+) ve) or anticlockwise (assuming (–) ve) direction.

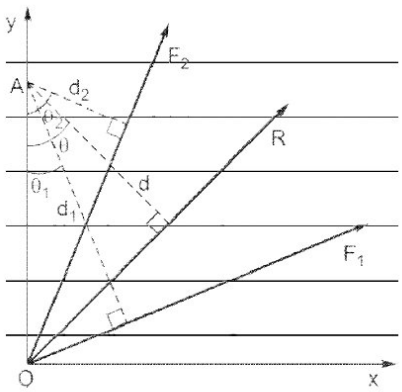


Point about which moment is calculated is called moment centre and the perpendicular distance of the force from the point about which moment is calculated is termed as moment arm.

VERIGNON'S THEOREM

It states that the algebraic sum of moments due to all forces acting on an object about any point is equal to the moment because of their resultant about the same point.

Considering two force F_1 and F_2 whose resultant is R. Angle from y-axis and perpendicular distances from point A, about which moment have to be calculated, are shown in Fig.



Moment due to resultant R about point A is

$$M_A = R \,.\, d_1 = R \,.\, (OA \cos \theta)$$
$$= OA\,(R \cos \theta) \qquad ...(1)$$

Moment due to force F_1 about point A is

$$M_{1A} = F_1 \,.\, d,$$
$$= F_1 \,.\, (OA \cos \theta_1)$$
$$= OA \,.\, (F_1 \cos \theta_1) \qquad ...(2)$$

Moment due to force F_2 about point A is

$$M_{2A} = F_2 \,.\, d_2 = F_2\,(OA \cos \theta_2)$$
$$= OA\,(F_2 \cos \theta_2) \qquad ...(3)$$

Adding equation (2) and (3)

$$M_{1A} + M_{2A} = OA\,[F_1 \cos \theta_1 + F_2 \cos \theta_2]$$

or $$M_{1A} + M_{2A} = M_A \qquad [\because \; F_1 \cos \theta_1 \text{ and } F_2 \cos \theta_2]$$

are the x-components of force F_1 and F_2 which must be equal to the x-component of resultant R *i.e.*, $R \cos \theta$.